

Handover Document: Development of an Open Source Autopilot for QUT

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1 Project Overview

This project involved refactoring, building, programming, and testing an open-source flight module design to serve as an in-house replacement for UAV units at QUT. The design is an evolution of the AUAV Pixracer (FMUv4 generation), licensed under Creative Commons Attribution-ShareAlike 3.0.

The work presented was derived from the v1 version produced by Wesley (KAI) in 2022.

2 Hardware Specifications

The autopilot is optimized for small racing quads and planes, supporting up to 6 PWM outputs.

2.1 Standard Specifications (v1)

- **MCU:** STM32F427VIT6 CPU (180 MHz ARM Cortex M4).
- **Connectivity:** In-built WiFi, JST GH connectors, microSD logging.
- **Sensors:** Invensense MPU9250 and ICM-20608, MS5611 Barometer, and HMC5983 Compass.

2.2 Modified Specifications (v2)

Modified sensors were integrated to modernize the board:

- **Gyro/Accelerometer:** Invensense IMC-42688-P and ICM-20608-G.
- **Compass:** LIS3MDLTR Mag.
- **New Features:** Added dedicated Reset and BOOT0 buttons.

3 Firmware and Software

The system utilizes **PX4 firmware** loaded via DFU.

- **Modifications:** Drivers and SPI chip select pins were modified for custom hardware compatibility.
- **Telemetry:** Mavlink 8266 firmware was loaded onto the external ESP-01 WiFi chip.
- **Testing:** Custom firmware was designed to test sensor communication, using the onboard LED to signal specific communication failures.

4 Project Timeline and Milestone Notes

The following log represents the development progress and testing phase:

- **11/10/2022:** Initial project proceeding.
- **01/06/2023:** Development continued following the initial refactor.
- **08/12/2025:** Finalized board configuration and functional requirements.
- **09/12/2025:** Validated critical pins, including PA4-7, PD15, and PC2.
- **23/12/2025:** Reviewed PCB layout metrics, specifically aspect ratio and solder-mask bridges.
- **05/01/2026:** Conducted a major design review for the next iteration(missing 3.3V to STM)
- **28/01/2026: Critical Failure.** 12.5V was sent into the board, causing it to catch fire.
- **02/02/2026: Fabrication Success.** V2 board was fabricated. Custom firmware was designed to test sensor communication, using the onboard LED to signal specific communication failures. All sensors were communicating properly.
- **03/02/2026: Firmware Success.** PX4 firmware was successfully loaded and mav8266 firmware was successfully loaded onto the ESP-01 WiFi chip.
- **10/02/2026: Connectivity Success.** Connectivity to QGroundControl was verified. Basic Airframe was assembled and tested for basic operations.
- **20/02/2026: Flight Test Success.** The drone was able to take off, hover and fly. Demonstrating basic stability in manual mode. This was done either using RC or MAVLink commands from QGroundControl.
- **24/02/2026: Companion Computer Integration.** The airframe was updated with a companion computer and used TELM1 and a usb-uart connection connection to the V2 board worked. The drones mavros node was able to collect position data and map using rviz2.

- **25/02/2026: Autonomy Test Partial Success.** Full autonomous flight was not achieved due to a error with offboarding mode, However position hold and position tracking using a companion computer and ros2 was successful.
- **10/03/2026: V3 Design Review.** Issues were identified for the next board revision, including labeling, thermal relief, component placement, USB routing, sensor interference, debugging access, and mechanical design. Handover document was finalized and submitted.

5 Flight Tests and Validation

At the end of the project, the prototype was prepared for flight tests to validate the design's stability.

- **Goal:** Test design validity and sensor accuracy under flight conditions.
- **Status:** Connectivity to QGroundControl was verified for basic operations on a simple airframe.
- **Stability:** The drone was able to take off, hover and fly. Demonstrating basic stability in manual mode. This was done either using RC or MAVLink commands from QGroundControl.
- **Autonomy:** Full autonomous flight was not achieved due to a error with offboarding mode, However a compainion computer was connected to the board and used TELM1 and a usb-uart connection connection to the V2 board worked. The drone used a inside camera based tracking system and ros2 to collect position data and preform hold position operation which were successful.

6 Current Issues and Improvements for V3

The following issues were identified for the next board revision:

1. **Power Supply:** The 3.3V line to the STM was missing, which is critical for board operation.
2. **Labeling:** D4 label is missing; overall board labels are too small to read.
3. **Thermal Relief:** No thermal relief on the copper polygons/ground pads.
4. **Component Placement:** Place all small components (STM, Sensors) on a single face to allow for easier solder paste mask and reflow.
5. **USB Routing:** USB data lines are not currently of equal length.
6. **Sensor Interference:** Rotate one of the accelerometer sensors 90 degrees so they do not share the same orientation.
7. **Debugging:** Add a breakout for serial debugging pins to avoid using DFU lines.
8. **Mechanical:** Reverse the SD card footprint for better accessibility; increase board size/mounting holes for better case integration.

7 Future Work

- Design an outer case with proper vibration isolation and barometer foam protection.
- Consider an Ethernet header or a Raspberry Pi header for non-WiFi modes.
- Finalize external production setup once V3 issues are resolved.